

Manual

Bug Report Classification Tool — User Guide

Overview

This tool classifies GitHub bug reports from deep learning framework projects as either **performance-related** (class 1) or **not performance-related** (class 0). It compares five classifiers — Naive Bayes (baseline), SVM, Random Forest, Logistic Regression, and XGBoost — across five open-source projects: TensorFlow, PyTorch, Keras, MXNet, and Caffe.

How to Run

From the `lab1/` directory, run:

```
python main.py
```

No command-line arguments are required. The script is fully self-contained and will:

- Load and preprocess all five project datasets from `datasets/`
- Run all five classifiers with 30 repeated random train/test splits per project
- Compute Precision, Recall, and F1 for each run
- Perform Wilcoxon signed-rank tests comparing each classifier against the baseline
- Save results to `results/` as CSV files
- Generate F1 box plot figures to `results/figures/`
- Print a summary table to the console

Expected Output Files

File	Description
<code>results/raw_results.csv</code>	All 750 individual run scores (5 classifiers x 5 projects x 30 runs)
<code>results/summary_results.csv</code>	Mean and std of Precision, Recall, F1 per classifier per project
<code>results/wilcoxon_tests.csv</code>	Wilcoxon p-values for each classifier vs baseline, per project
<code>results/figures/*_f1_boxplot.png</code>	Box plot of F1 distributions for each project

Expected Runtime

Approximately 2-5 minutes on a modern laptop. The majority of time is spent on Random Forest and XGBoost training across the 30 repeats.

Console Output

The script prints progress updates as it processes each project, followed by a formatted summary table showing mean F1 scores and Wilcoxon test significance for all classifiers.